Number Representation & Arithmetic Operations

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Outline

- **Positional Number Representation**
  - Review of Decimal Numbers
  - Binary, Octal, Hexadecimal
  - Conversion between different representations
- **Non-positional Number Representation**
  - Roman Numeral System
- **Number Representation in Computer**
  - Signed numbers
  - Signed negative numbers
    - Sign-and-magnitude
    - 1's complement
    - 2's complement
    - Addition and subtraction
  - Real number
    - Fixed-point numbers
    - Floating-point numbers

Review of Decimal Numbers

- **Review of Decimal (十进制)**

positive integer	**11235813**
negative integer	**- 11235813**
positive real number	**11235813.2134**
negative real number	**- 11235813.2134**

$$D = d_{n-1}d_{n-2}\cdots d_1 d_0 . d_{-1}d_{-2}\cdots d_{-k}$$

$$V(D) = d_{n-1}\times 10^{n-1} + d_{n-2}\times 10^{n-2} + \cdots + d_0 \times 10^0 + d_{-1}\times 10^{-1} + \cdots d_{-k}\times 10^{-k} = \sum_{i=-k}^{n-1} d_i \times 10^i$$

Example: $11235813.2134 = 1\times 10^7 + 1\times 10^6 + \cdots + 3\times 10^0 + 2\times 10^{-1} + \cdots 4\times 10^{-4}$

Positional Number Representation

- Positional number representation **(按位记数法)**
- A method of representing numbers where the value of each digit depends on its **position** 位置决定数值

$$X = \pm x_{n-1}x_{n-2}\cdots x_1x_0 \cdot x_{-1}x_{-2}\cdots x_{-k}$$

$$V(X) = \pm\left(x_{n-1}\times r^{n-1} + \cdots + x_0 \times r^0 + x_{-1}\times r^{-1} + \cdots x_{-k}\times r^{-k}\right) = \pm\sum_{i=-k}^{n-1} x_i \times r^i$$

$$r : radix \ \text{(基数)}$$

Positional Number Representation

- **Binary (二进制)**
  Digits: **0 1**
- **Octal (八进制)**
  Digits: **0 1 2 3 4 5 6 7**
- **Hexadecimal (十六进制)**
  Digits: **0 1 2 3 4 5 6 7 8 9 A B C D E F**

$$(1\ 1)_2 = 2^1 + 2^0 = (3)_{10}$$

$$(1\ 1)_8 = 8^1 + 8^0 = (9)_{10}$$

$$(1\ 1)_{16} = 16^1 + 16^0 = (17)_{10}$$

Decimal	Binary	Octal	Hexadecimal
00	00000	00	00
01	00001	01	01
02	00010	02	02
03	00011	03	03
04	00100	04	04
05	00101	05	05
06	00110	06	06
07	00111	07	07
08	01000	10	08
09	01001	11	09
10	01010	12	0A
11	01011	13	0B
12	01100	14	0C
13	01101	15	0D
14	01110	16	0E
15	01111	17	0F
16	10000	20	10
17	10001	21	11
18	10010	22	12

Conversion Between Different Representations

X → decimal

$$X = \pm x_{n-1}x_{n-2}\cdots x_1x_0 \cdot x_{-1}x_{-2}\cdots x_{-k}$$

$$V(X) = \pm\left(x_{n-1}\times r^{n-1} + \cdots + x_0 \times r^0 + x_{-1}\times r^{-1} + \cdots x_{-k}\times r^{-k}\right) = \pm\sum_{i=-k}^{n-1} x_i \times r^i$$

$$(1\ 1)_2 = 2^1 + 2^0 = (3)_{10}$$

$$(1\ 1)_8 = 8^1 + 8^0 = (9)_{10}$$

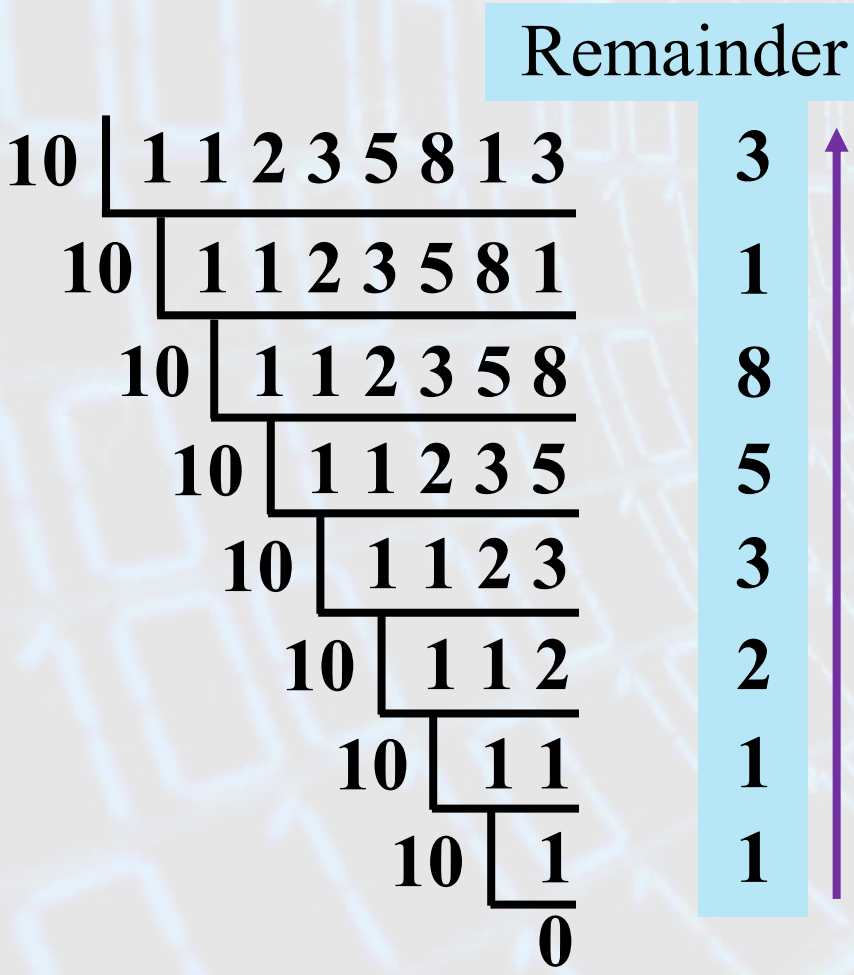
$$(1\ 1)_{16} = 16^1 + 16^0 = (17)_{10}$$

$$(\mathrm{F\ C})_{16} = 15\times16^1 + 12\times16^0 = (252)_{10}$$

Conversion Between Different Representations

decimal → X

Example: decimal **integer**: **1 1 2 3 5 8 1 3**



LSB: least significant bit

- ✓ divide by 10 successively
- ✓ record remainders
- ✓ get remainders from MSB to LSB

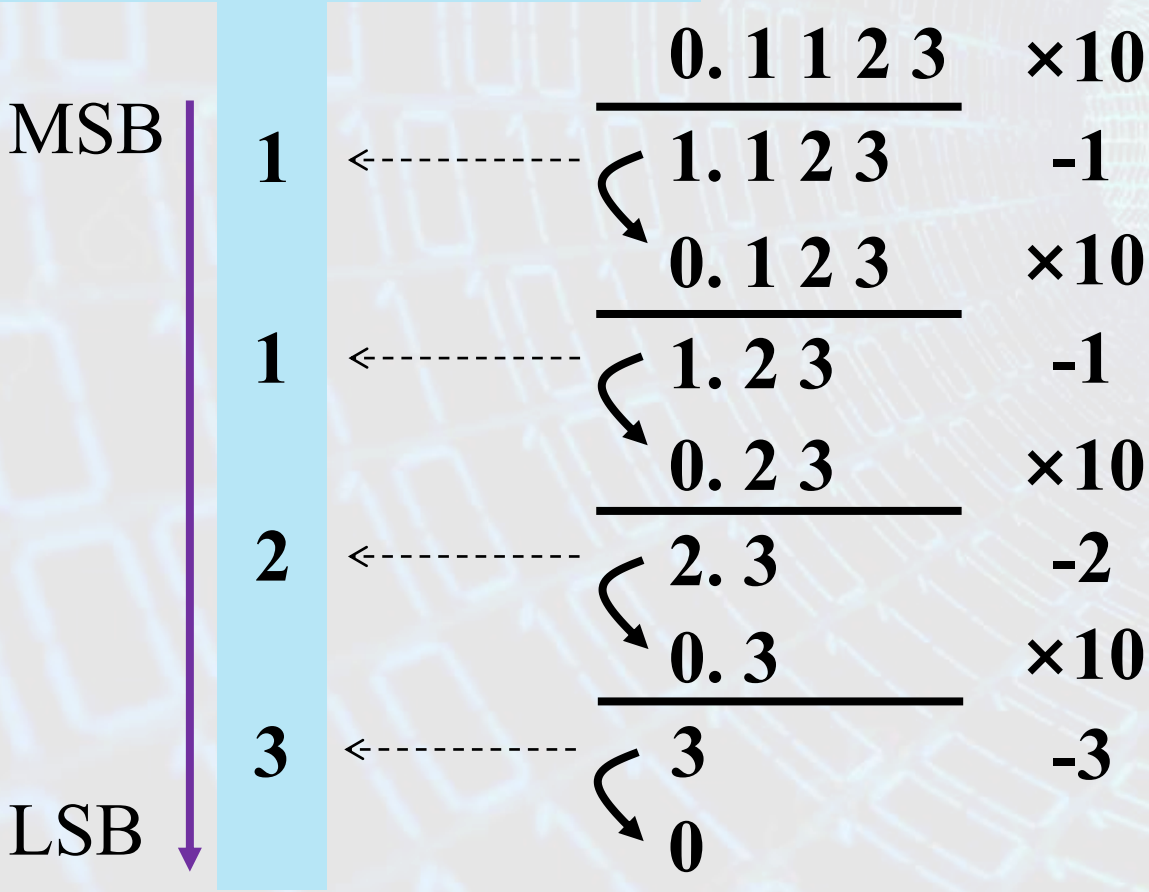
MSB: most significant bit

Conversion Between Different Representations

decimal → X

Example: decimal **fraction**(小数): **0. 1 1 2 3**

Integer part (整数部分)



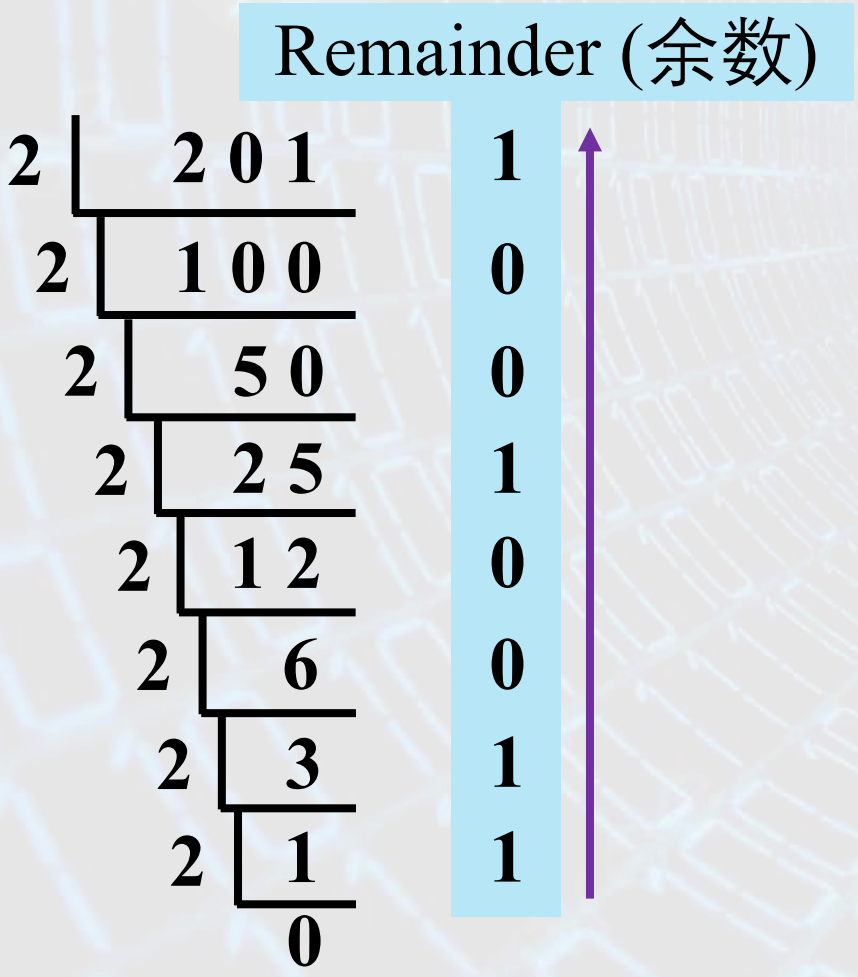
- ✓ multiply by 10 successively
- ✓ subtract integer part and record
- ✓ get integer part from MSB to LSB

Conversion Between Different Representations

decimal → binary

Example:

convert decimal number $(2\ 0\ 1)_{10}$ to binary number



$$(2\ 0\ 1)_{10} = (1\ 1\ 0\ 0\ 1\ 0\ 0\ 1)_2$$

Conversion Between Different Representations

decimal → binary

Example:
convert decimal number $(0.8254)_{10}$ to binary number

Integer part (整数部分)

MSB

1	←	0.8254 ×2
1	←	1.6508 ×2
0	←	1.3016 ×2
1	←	0.6032 ×2
0	←	1.2064 ×2
0	←	0.4128 ×2
0	←	0.8256 ×2
1	←	0.6512 ×2
1	←	1.3024 ×2
⋮		⋮

LSB

$$(0.8254)_{10} = (0.11010011\cdots)_2$$

Conversion Between Different Representations

decimal → hexadecimal

Example: convert decimal number $(2\ 0\ 1)_{10}$ to hexadecimal number

Remainder (余数)

16	2 0 1	9
16	1 2	C
	0	

$$(2\ 0\ 1)_{10} = (C\ 9)_{16}$$

Conversion Between Different Representations

binary → octal

$8 = 2^3$

Take 3 bits as one group

integer

$$\underline{0\,0}\;\underline{1\,0\,1}\;\underline{0\,1\,1}\;\underline{1\,0\,0}\;\underline{1\,0\,0}\;\underline{1\,0\,1}$$

$$1\quad 2\quad 7\quad 4\quad 4\quad 5$$

$$(101011100100101)_2 = (127445)_8$$

fraction

$$0.\underline{1\,0\,1}\;\underline{0\,1\,1}\;\underline{1\,0\,1}$$

$$5\quad 3\quad 5$$

$$(0.10101110)_2 = (0.535)_8$$

binary → hexadecimal

$16 = 2^4$

Take 4 bits as one group

integer

$$\underline{1\,0\,1\,0}\;\underline{1\,1\,1\,1}\;\underline{0\,0\,1\,0}\;\underline{0\,1\,0\,1}$$

$$A\quad F\quad 2\quad 5$$

$$(101011100100101)_2 = (AF25)_{16}$$

fraction

$$0.\underline{1\,0\,1\,0}\;\underline{1\,1\,1\,0}\;\underline{1\,0\,0\,0}$$

$$A\quad E\quad 8$$

$$(0.10101110)_2 = (0.AE8)_{16}$$

Non-positional Number Representation

- **Roman Numeral System**

primary symbols

Roman	Arabic
I	1
V	5
X	10
L	50
C	100
D	500
M	1000

- A symbol placed after another equal or greater value adds its value
  e.g. $II = 1 + 1 = 2$, $LVIII = 50 + 5 + 1 + 1 + 1 = 58$
- A symbol is placed before one of greater value to subtract its value
  e.g. $IV = -1 + 5 = 4$, $LIX = 50 - 1 + 10 = 59$
- A bar (vinculum) placed over a number multiplies its value by 1,000.
  e.g. $\overline{V} = 5 \times 1000 = 5000$
- Usually only three identical symbols can be used consecutively;
- 4,000 is sometimes written as MMMM, and clocks usually display 4 as IIII.

https://www.rapidtables.com/math/symbols/roman_numerals.html

Non-positional Number Representation

- **Roman Numeral System**

Roman	Calculation	Arabic
III	1+1+1	3
IV	5-1	4
VIII	5+1+1+1	8
XVIII	10+5+1+1+1	18
XIX	10+(10-1)	19
XLIX	(50-10)+(10-1)	49
LXXII	50+10+10+1+1	72
MDC	1000+500+100	1600
MMVII	1000+1000+5+1+1	2007
MMXXV	1000+1000+10+10+5	2025
……	……	……

https://www.rapidtables.com/math/symbols/roman_numerals.html

Number Representation in Computer

positive integer	11235813
negative integer	-11235813
positive real number	11235813.2134
negative real number	-1123581.32134

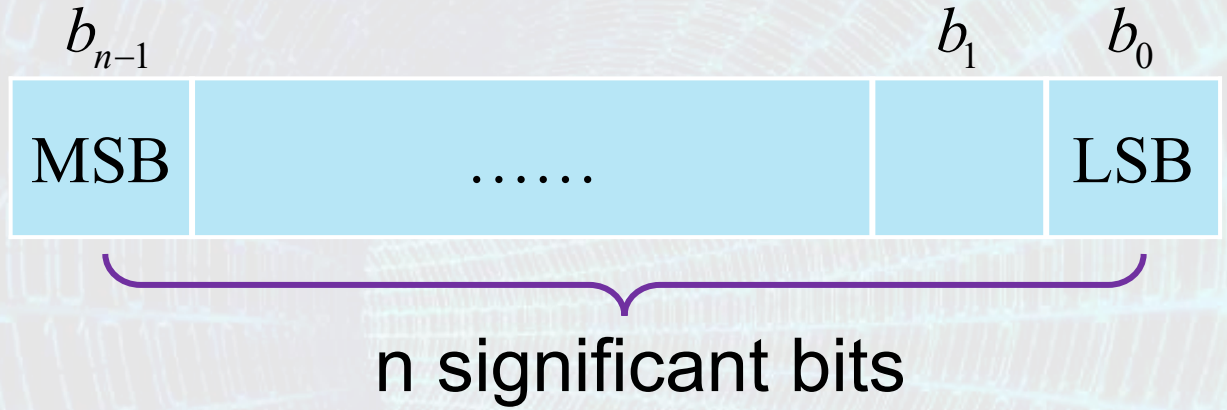


Question: How to store negative and real numbers ?

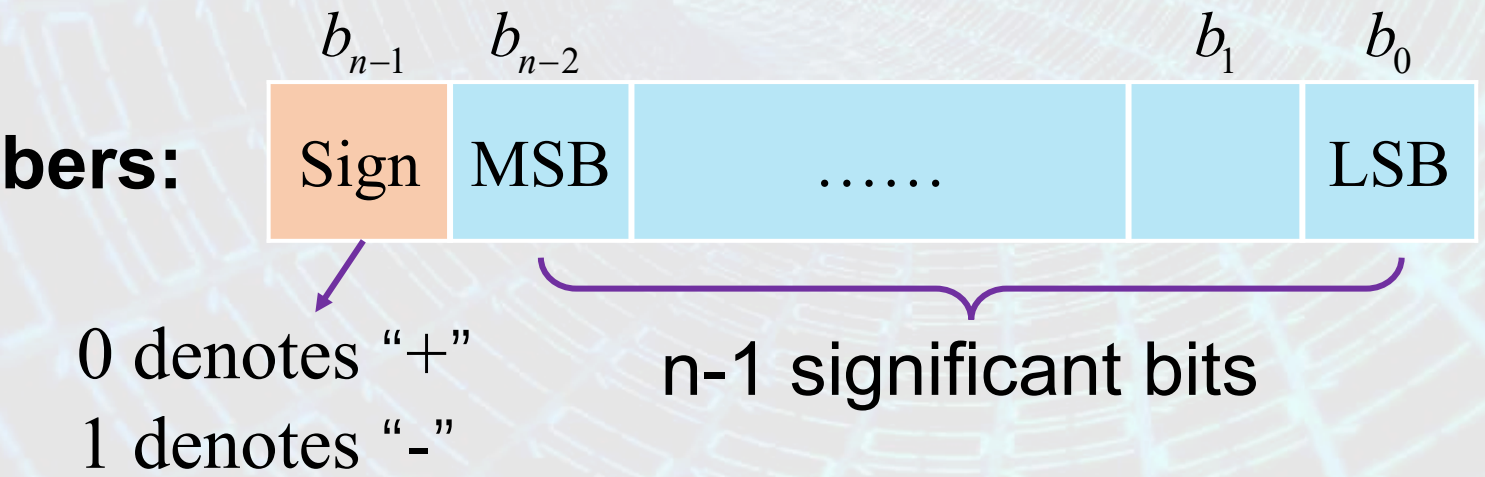
- *Store a character "-" or "." ?*
- *Use more bits for "-" or "." ?*
- *How does the "-" "." used in arithmetic operation?*

Signed Numbers (有符号数)

Unsigned numbers:

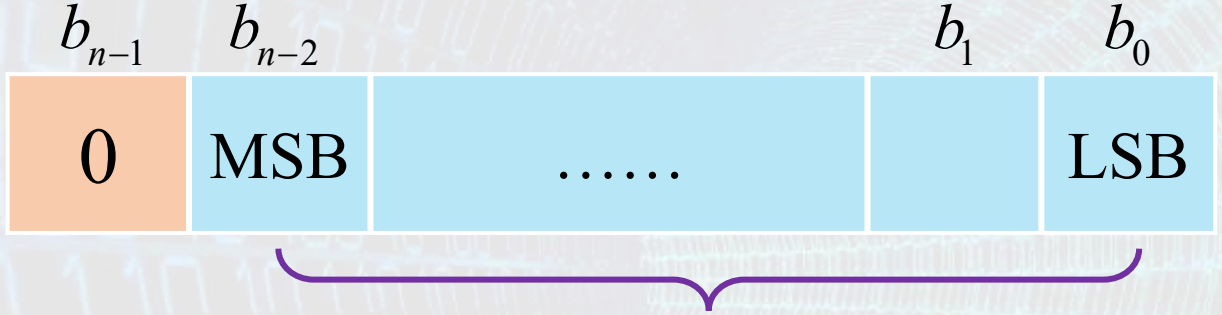


Signed numbers:



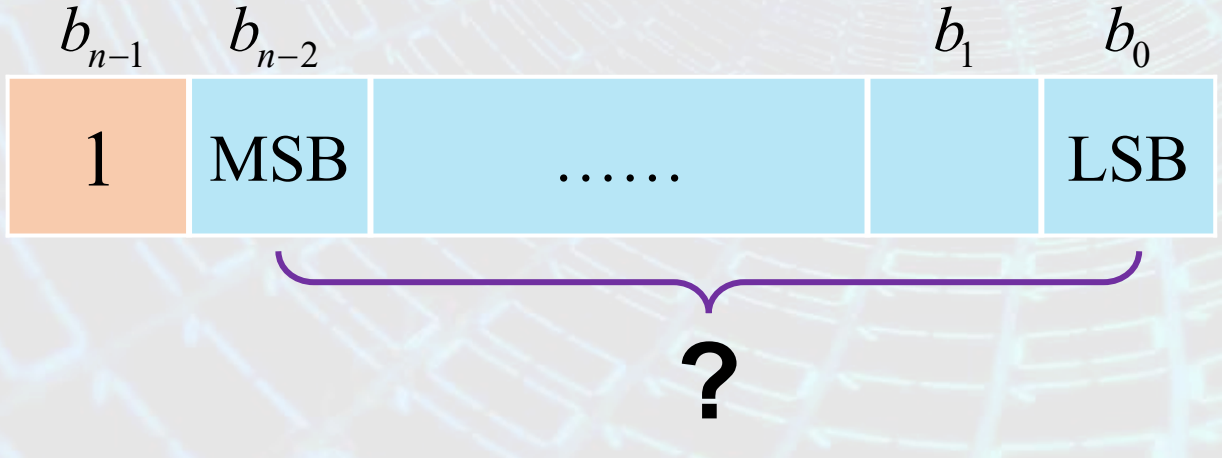
Signed Integers

- **Signed positive numbers:**



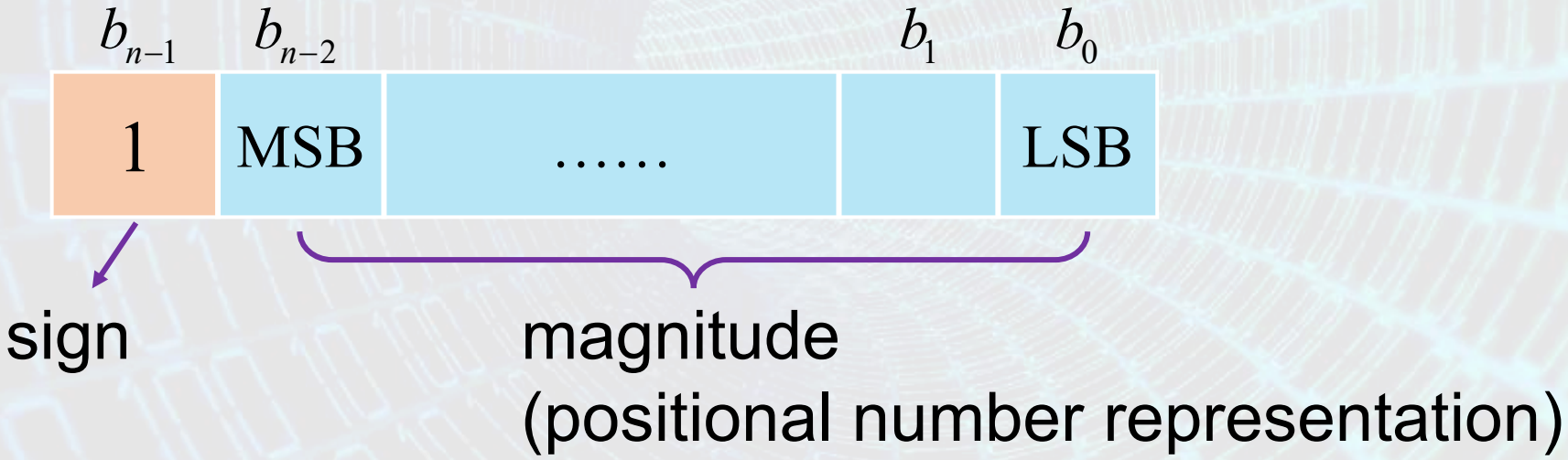
magnitude (positional number representation)

- **Signed negative numbers:**



Signed Negative Integers

- **Sign-and-magnitude representation**
  (符号-模值表示法 或 原码表示法)



e.g. $(+5)_{10} = (0\ 1\ 0\ 1)_2$

$(-5)_{10} = (1\ 1\ 0\ 1)_2$

Signed Negative Integers

- **1's complement representation** (1的补码表示法):

Method 1 $K_1 = (2^n - 1) - P$

K_1 : 1's complement representation

P : equivalent positive number $P = (+5)_{10} = (0\ 1\ 0\ 1)_2$

e.g. $(-5)_{10} = (2^4 - 1) - (0\ 1\ 0\ 1)_2 = (1\ 1\ 1\ 1)_2 - (0\ 1\ 0\ 1)_2 = (1\ 0\ 1\ 0)_2$

Method 2 Complementing each bit of P, including the sign bit

complement ⤹

$(+5)_{10} = (0\ 1\ 0\ 1)_2$

$(-5)_{10} = (1\ 0\ 1\ 0)_2$

Signed Negative Integers

- **2's complement representation** (2的补码表示法):

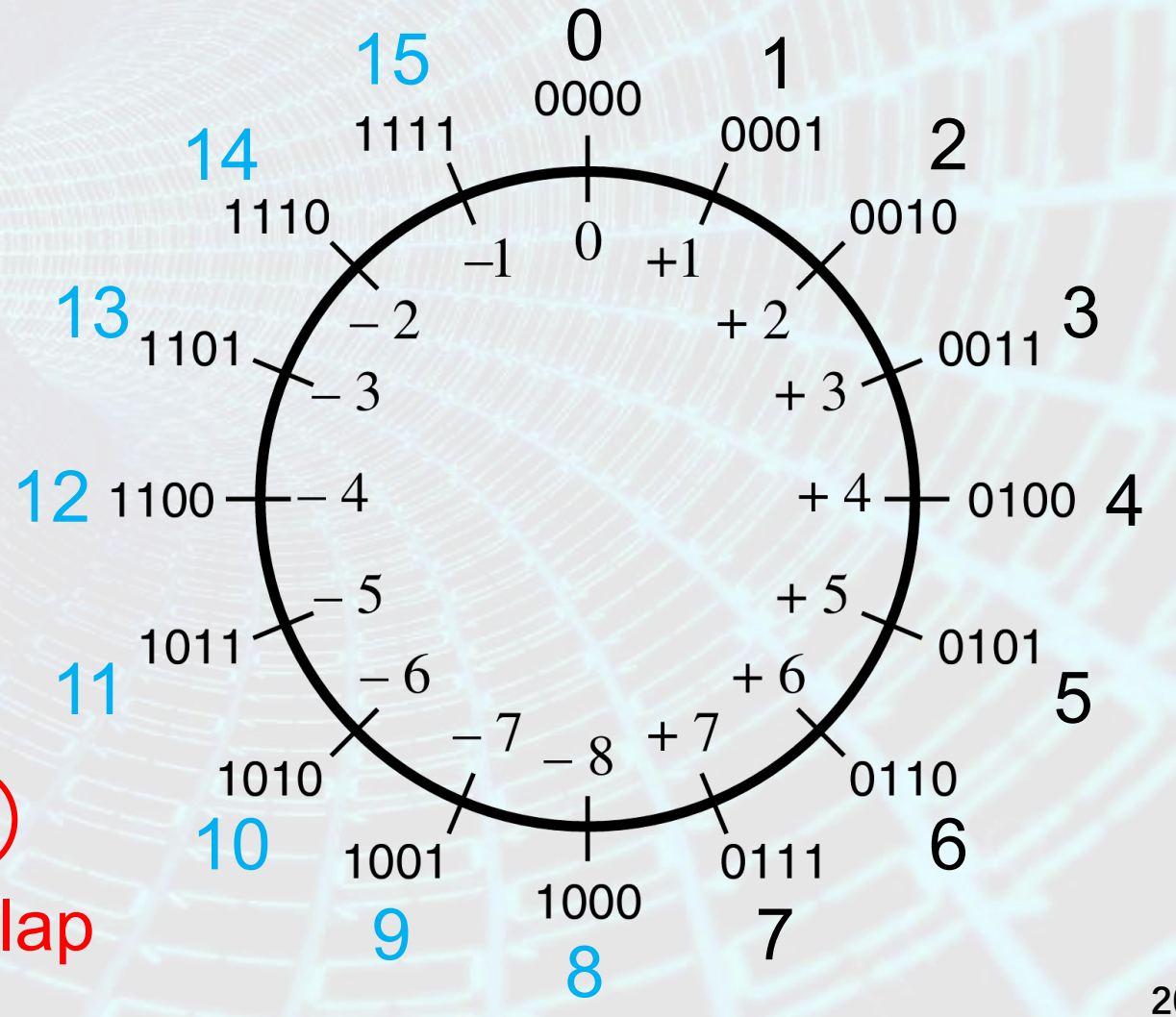
$K_2 = 2^n - P \quad or \quad K_2 = K_1 + 1$

K_2 : 2's complement representation

P : equivalent positive number

e.g. $(-5)_{10} = 2^4 - 5 = (11)_{10} = (1\ 0\ 1\ 1)_2$

$(-5)_{10} = (1\ 0\ 1\ 0)_2 + 1 = (1\ 0\ 1\ 1)_2$



$K_2 = 2^n - P \rightarrow K_2 + P = 2^n$

one lap

Signed Negative Integers

$b_3b_2b_1b_0$	Sign and magnitude	1's complement	2's complement
0111	+7	+7	+7
0110	+6	+6	+6
0101	+5	+5	+5
0100	+4	+4	+4
0011	+3	+3	+3
0010	+2	+2	+2
0001	+1	+1	+1
0000	+0	+0	+0
1000	−0	−7	−8
1001	−1	−6	−7
1010	−2	−5	−6
1011	−3	−4	−5
1100	−4	−3	−4
1101	−5	−2	−3
1110	−6	−1	−2
1111	−7	−0	−1

Same for positive integers

Issue: two representation for "0"

2's complement scheme is better

Practice:

Problem: Determine the decimal values of the following 8 bits when it's stored as:

① unsigned integer

② signed integer with sign-and-magnitude scheme

③ signed integer with 1's complement scheme

④ signed integer with 2's complement scheme

1	0	1	1	0	1	1	1

Solution: ① as unsigned integer:

$$V = 2^7 + 2^5 + 2^4 + 2^2 + 2^1 + 2^0 = (183)_{10}$$

Practice:

1	0	1	1	0	1	1	1

Solution: ② as signed integer with sign-and-magnitude scheme :

$$V = -\left(2^5 + 2^4 + 2^2 + 2^1 + 2^0\right) = \left(-55\right)_{10}$$

③ as signed integer with 1's complement scheme :

Magnitude:	0	1	0	0	1	0	0	0

$$V = -\left(2^6 + 2^3\right) = \left(-72\right)_{10}$$

④ as signed integer with 2's complement scheme :

Magnitude:	0	1	0	0	1	0	0	1

$$V = -\left(2^6 + 2^3 + 2^0\right) = -\left(73\right)_{10}$$

Addition(加法) of Signed Integers

	sign-and-magnitude	1's complement	2's complement
(+5)	0 1 0 1	0 1 0 1	0 1 0 1
+ (+2)	+ 0 0 1 0	+ 0 0 1 0	+ 0 0 1 0
(+7)	0 1 1 1	0 1 1 1	0 1 1 1
	$(+7)_{SM}$=0 1 1 1 ✓	$(+7)_{1's}$=0 1 1 1 ✓	$(+7)_{2's}$=0 1 1 1 ✓
(–5)	1 1 0 1	1 0 1 0	1 0 1 1
+ (–2)	+ 1 0 1 0	+ 1 1 0 1	+ 1 1 1 0
(–7)	1 0 1 1 1	1 0 1 1 1	1 1 0 0 1
	$(-7)_{SM}$=1 1 1 1 ✗	$(-7)_{1's}$=1 0 0 0 ✗	$(-7)_{2's}$=1 0 0 1 ✓

Addition of Signed Integers

	sign-and-magnitude	1's complement	2's complement
(–5)	1 1 0 1	1 0 1 0	1 0 1 1
+ (+2)	+ 0 0 1 0	+ 0 0 1 0	+ 0 0 1 0
(–3)	1 1 1 1	1 1 0 0	1 1 0 1
	$(-3)_{SM}$=1 0 1 1 ✗	$(-3)_{1's}$=1 1 0 0 ✓	$(-3)_{2's}$=1 1 0 1 ✓
(+5)	0 1 0 1	0 1 0 1	0 1 0 1
+ (–2)	+ 1 0 1 0	+ 1 1 0 1	+ 1 1 1 0
(+3)	1 1 1 1	1 0 0 1 0	1 0 0 1 1
	$(+3)_{SM}$=0 0 1 1 ✗	$(+3)_{1's}$=0 0 1 1 ✗	$(+3)_{2's}$=0 0 1 1 ✓

Subtraction(减法) of Signed Integers

- Subtraction can be performed by addition
- Negate(取反) the subtrahend(减数) and add it to the minuend(被减数)

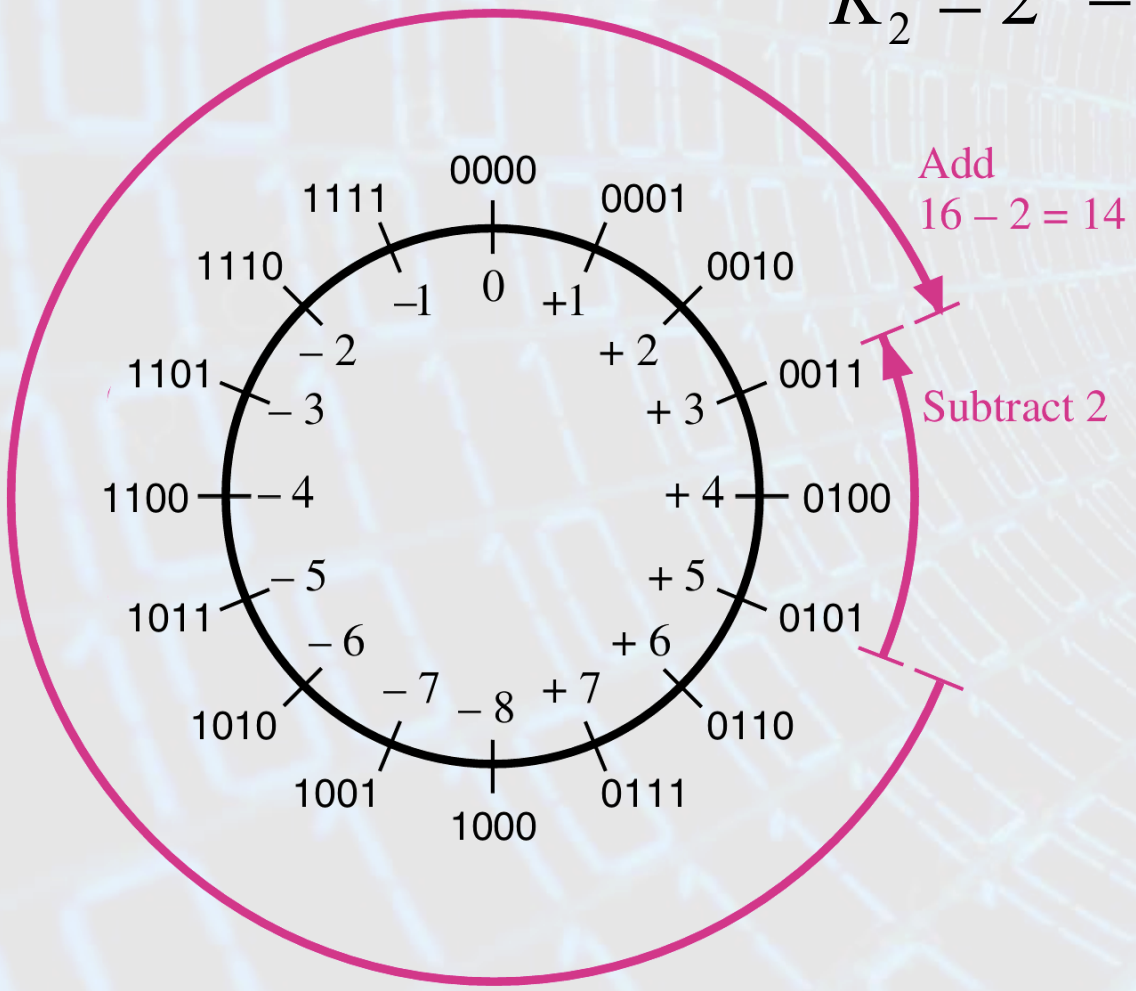
$$A - B = A + (-B)$$

(+5)	(+5)	(–5)	(–5)
– (+2)	– (–2)	– (+2)	– (–2)
(+3)	(+7)	(–7)	(–3)
↓	↓	↓	↓
(+5)	(+5)	(–5)	(–5)
+ (–2)	+ (+2)	+ (–2)	+ (+2)
(+3)	(+7)	(–7)	(–3)

Subtraction of Signed Integers

- **Why is 2's complement suitable for subtraction**

$$K_2 = 2^n - P \quad \Rightarrow \quad K_2 + P = 2^n$$



Subtracting 2 by adding its 2's complement

Radix-Complement Schemes

$$K_1 = (2^n - 1) - P$$
$$K_2 = 2^n - P$$

- **Take decimal as example**

  How do we compute **74 - 36 ?**

$$\begin{array}{r} {}^{1} \\ 7\ 4 \\ -\ 3\ 6 \\ \hline 3\ 8 \end{array}$$

borrow(借位)

$$\begin{array}{r} (+5) \\ +\ (-2) \\ \hline (+3) \end{array} \quad \rightarrow \quad \begin{array}{r} 0\ 1\ 0\ 1 \\ +\ 1\ 1\ 1\ 0 \\ \hline 1\ 0\ 0\ 1\ 1 \end{array}$$

$(+3)_{2's} = 0\ 0\ 1\ 1$



ignore the carry

An easier way is:

$$74 - 36 = 74 + (-36) + 100 - 100$$
$$= 74 + 10^2 - 36 - 100$$

10's complement of 36

$$= 74 + 99 - 36 + 1 - 100$$

9's complement of 36

Addition and Subtraction

Summary

- Sign-and-magnitude scheme cannot deal with addition of negative number
- 1's complement scheme cannot deal with addition of negative addend (加数)
- 2's complement scheme does addition correctly

- Subtraction can be performed by addition
- 2's complement scheme does both addition and subtraction correctly

Real Numbers



Question: How to store negative and real numbers ?

- **Fixed-Point Numbers**

$$X = \pm x_{n-1}x_{n-2}\cdots x_1x_0 \cdot x_{-1}x_{-2}\cdots x_{-k}$$

The position of the radix point (小数点) is assumed to be fixed

$(27.5)_{10}$

0	1	1	0	1	1	1	0

$(-18.25)_{10}$

1	1	0	0	1	0	0	1

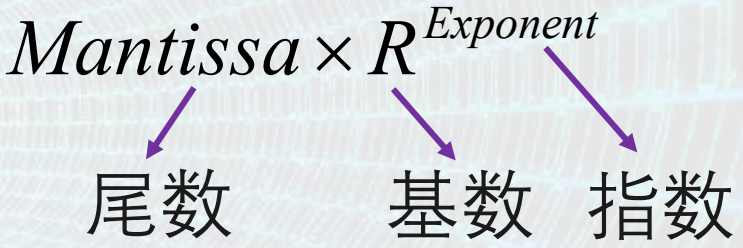
Issue: limited bits leads to limited range of number

Real Numbers

- **Floating-Point Numbers**

$2.75432 \times 10^1 = 27.5432$
$2.75432 \times 10^2 = 275.432$
$2.75432 \times 10^3 = 2754.32$
$2.75432 \times 10^4 = 27543.2$

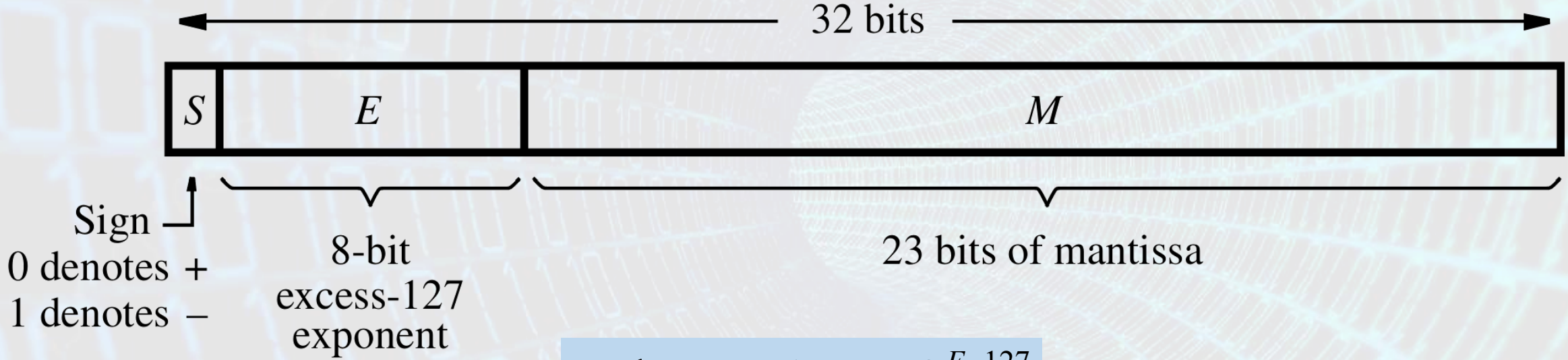
The position of the radix point is changing



- **Two format for binary floating-point numbers**
  - IEEE 754 standard
  - Single-precision 32-bit floating-point format
  - Double-precision 64-bit floating-point format

Real Numbers

- **Single-precision format**



$$\text{Value} = \pm 1.M \times 2^{E-127}$$

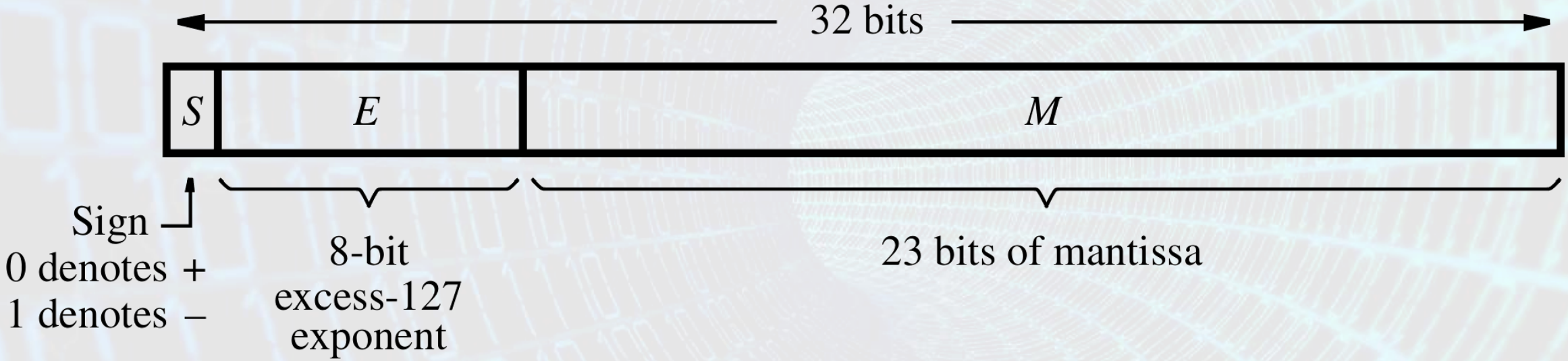
the most-significant bit of mantissa is always equal to “1”, so not represented

$$E \in [0, 255]$$

$$\begin{cases} if\ E = 0, \text{Value} = 0 \\ if\ E = 255, \text{Value} = +\infty \\ else: Exponent = E - 127 \end{cases}$$

Real Numbers

- **Single-precision format**



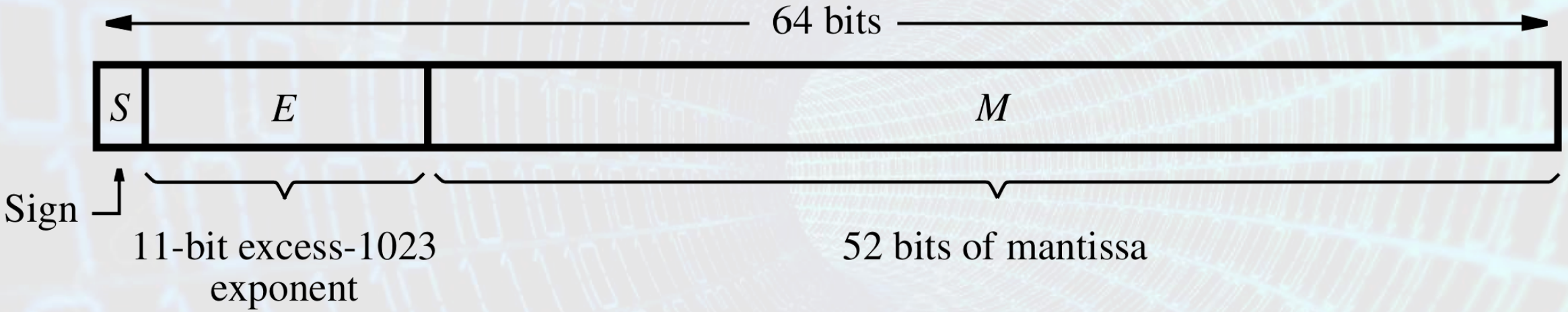
Largest absolute value: $1.11\ldots1\times2^{254-127}\approx3.4\times10^{38}$

Smallest absolute value: $1.00\ldots0\times2^{1-127}\approx1.18\times10^{-38}$

Total range: $\left[-3.4\times10^{38},-1.18\times10^{-38}\right]\cup\left[1.18\times10^{-38},3.4\times10^{38}\right]$

Real Numbers

- **Double-precision format**



$$\text{Value} = \pm 1.M \times 2^{E-1023}$$

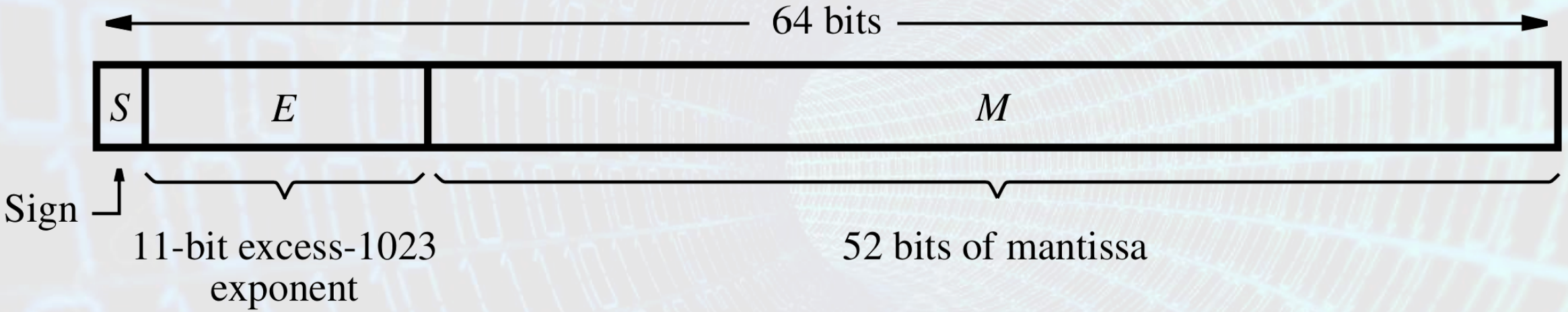
the most-significant bit of mantissa is always equal to “1”, so not represented

$$E \in [0, 2047]$$

$$\begin{cases} if\ E = 0, \text{Value} = 0 \\ if\ E = 2047, \text{Value} = +\infty \\ else: Exponent = E - 1023 \end{cases}$$

Real Numbers

- **Double-precision format**



Largest absolute value: $1.11\ldots1\times2^{2046-1023}\approx1.8\times10^{308}$

Smallest absolute value: $1.00\ldots0\times2^{1-1023}\approx2.23\times10^{-308}$

Total range: $\left[-1.8\times10^{308},-2.23\times10^{-308}\right]\cup\left[2.23\times10^{-308},1.8\times10^{308}\right]$

https://zhuanlan.zhihu.com/p/343037540

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